

The Hierarchy ν/E_p

Two Derivations Compared: FFGFT and Observer Patch Holography

Johann Pascher
ORCID 0009-0000-6518-4064

22 September 2026

Contents

The Hierarchy v/E_p : Two Derivations Compared	2
List of symbols	2
1 Occasion	2
2 The FFGFT derivation	3
3 The OPH derivation [Q]	3
4 Comparison	3
5 Entry for the comparison table	4
Tool and method	4
6 Result	4

The Hierarchy ν/E_P : Two Derivations Compared

Abstract. FFGFT obtains the electroweak scale from ξ with m_e as anchor (R104) and reaches the Planck scale through its own chain $\xi \rightarrow G \rightarrow E_P$, whose SI representation contains the measured value of G via C_{conv} . This route to the hierarchy number ν/E_P thus uses two measured values. Without an anchor, $\nu/E_P = \xi^4/(5\pi) = 2.0120 \times 10^{-17}$ (-0.23% , Doc. 149); the factor 10 in this formula is fitted to the measured value (Doc. 387). Observer Patch Holography (OPH) delivers the same number conditionally through its Theorem A. The two derivations give 2.0389×10^{-17} (FFGFT, $+1.10\%$) and 2.0200×10^{-17} (OPH, $+0.16\%$). The FFGFT deviation is entirely the known bare gap of the lepton ladder (Doc. 352); the Planck chain contributes less than 10^{-4} . The FFGFT entry for a hashed comparison table is given.

List of symbols

- [B] algebraically proven
- [K] numerically confirmed
- [S] structurally plausible, derivation open
- [Q] taken from an external source, cited

Symbol	Meaning
ξ	FFGFT scaling parameter (4/30000)
ν	electroweak scale
E_P	Planck energy (not reduced)
$G, C_{\text{conv}}, K_{\text{frak}}$	gravitational constant and factors of the SI representation (Docs. 012/013)
E_*, P, α_U	OPH reference energy, pixel ratio, unified coupling

1 Occasion

Doc. 374 brought FFGFT and OPH together at the electroweak scale and, in doing so, combined m_e/ν from the FFGFT ladder with ν/E_* from OPH. That understates the FFGFT side: FFGFT obtains ν itself, with m_e as anchor (R104), and reaches E_P through its own chain (Doc. 374, § 5). The hierarchy number ν/E_P is therefore not a combination but a second, independent derivation of the same quantity. This document places the two derivations side by side.

2 The FFGFT derivation

1. Electroweak scale (R104, Doc. 006):

$$\nu = \frac{m_e}{\frac{4}{3} \xi^{3/2}} = 248.93 \text{ GeV.} \quad (1)$$

2. Planck energy through its own chain (Docs. 012, 013, 180): $G = \xi^2/(4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$, hence $E_P = 1.22087 \times 10^{19} \text{ GeV}$.
3. Hierarchy:

$$\left. \frac{\nu}{E_P} \right|_{\text{FFGFT}} = 2.0389 \times 10^{-17} \quad \text{[K].} \quad (2)$$

Against the SM value $\nu = 246.22 \text{ GeV}$ (definition from G_F , R112) this is +1.10 % higher. The deviation is entirely that of ν ; the Planck chain contributes less than 10^{-4} . It is identical to the bare gap of the lepton ladder m_e/ν (−1.09 %, Doc. 352): ν is obtained from the bare relation and inherits its residual. Not a new finding.

This route uses two measured values: m_e for ν and, via C_{conv} , the measured value of G for E_P . Directly and without an anchor, ν/E_P follows from Doc. 149: $\nu/E_P = \xi^4/(5\pi) = 2.0120 \times 10^{-17}$, −0.23 %; the factor 10 in this formula is fitted to the measured value (Doc. 387). The factor 1.0134 between the two values is the residue of the electron mass in the chain (Doc. 383).

3 The OPH derivation [Q]

Theorem A from code/particles/hierarchy/theorem_package.md:

$$\frac{\nu}{E_\star} = P^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right] = 2.0200 \times 10^{-17}, \quad (3)$$

reproduced in the check script. The derivation is explicitly conditional in OPH; the identification $E_\star = E_P$ is a reading [S] (Doc. 374). Against the SM value: +0.16 %.

4 Comparison

Derivation	ν/E_P	vs. SM	Origin of the deviation
FFGFT	2.0389×10^{-17}	+1.10 %	bare residual of the ladder (Doc. 352)
OPH [Q]	2.0200×10^{-17}	+0.16 %	conditional fixed point, reading $E_\star = E_P$
SM comparator	2.0167×10^{-17}	—	ν from G_F (R112)

The two derivations lie 0.93 % apart. They also differ in sensitivity:

- FFGFT: $\partial \ln \nu / \partial \ln \xi = -3/2$ [B]. A rational structural quantity with power 3/2; small changes of ξ act weakly.
- OPH: $\partial \ln(\nu/E_\star) / \partial \alpha_U = \pi/(2\alpha_U^2) \approx 929$. The coupling sits in the exponent; the 0.16 % corresponds to a shift of α_U by 4×10^{-5} relatively.

The FFGFT value is robust and carries a known residual; the OPH value is more accurate but highly sensitive and conditional. Both statements are testable: on the FFGFT side through the resolution of the bare residual (Doc. 352), on the OPH side through the clock attachment and the identification of $E_\star$.

5 Entry for the comparison table

For a hashed comparison table the FFGFT entry, including the route, reads:

```
v/E_P | FF6FT | 2.03894e-17 | route: v=m_e/((4/3)xi^(3/2)) (R104, Doc 006) ;
E_P via G=xi^2/(4m_e)C_conv K_frak (Docs 012/013/180) ; bare ladder
```

SHA-256 of the string (generated in the check script):

```
84fac85fc55d8c5928a07550e774683961613059415b0ef02715b327f58fb15f
```

Tool and method

Claude (Anthropic) was used as a tool for: writing and running the Python check script, drafting the $\text{\LaTeX}$ sources, and maintaining the changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. OPH values are taken verbatim from the public repository; measured values (CODATA 2022, PDG) serve only as comparators.

6 Result

1. FFGFT delivers $v/E_p = 2.0389 \times 10^{-17}$ through its own chain: v from ξ with m_e as anchor (R104), E_p via G with the measured value of G in C_{conv} **[K]**. Without an anchor: $\xi^4/(5\pi) = 2.0120 \times 10^{-17}$ (-0.23%), factor 10 fitted to the measured value (Doc. 387).
2. The deviation of $+1.10\%$ is the bare residual of the lepton ladder; the Planck chain contributes less than 10^{-4} **[K]**.
3. OPH delivers, conditionally, 2.0200×10^{-17} ($+0.16\%$) **[Q]**; the derivations lie 0.93% apart.
4. Sensitivity: FFGFT $-3/2$ in $\ln \xi$ **[B]**, OPH ≈ 929 in α_U .
5. FFGFT table entry given with route and SHA-256.

Check script: 2/python/Dok375_Skripte/pruef_375_hierarchie.py (11/11)

Register (Doc. 190): no entry.

Bibliography

- [1] J. Pascher, *Doc. 006 — T_0 particle masses* (2025).
- [2] J. Pascher, *Doc. 012 — T_0 gravitational constant* (2025).
- [3] J. Pascher, *Doc. 013 — T_0 and SI units* (2025).
- [4] J. Pascher, *Doc. 180 — Derivation of L_0* (2026).
- [5] J. Pascher, *Doc. 352 — Lepton residuals* (2026).
- [6] J. Pascher, *Doc. 374 — Common anchor* (2026).
- [7] B. Mueller et al., *Observer Patch Holography*, github.com/FloatingPragma/observer-patch-holography (as of 22 September 2026).